

RESEARCH

Long-Term Depletion of Calcium and Other Nutrients in Eastern US Forests

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ABSTRACT / Both harvest removal and leaching losses can deplete nutrient capital in forests, but their combined long-term effects have not been assessed previously. We estimated changes in total soil and biomass N, Ca, K, Mg, and

P over 120 years from published data for a spruce–fir site in Maine, two northern hardwood sites in New Hampshire, central hardwood sites in Connecticut and Tennessee, and a loblolly pine site in Tennessee. For N, atmospheric inputs counterbalance the outputs, and there is little long-term change on most sites. For K, Mg, and P, the total pool may decrease by 2%–10% in 120 years depending on site and harvest intensity. For Ca, net leaching loss is 4–16 kg/ha/yr in mature forests, and whole-tree harvest removes 200–1100 kg/ha. Such leaching loss and harvest removal could reduce total soil and biomass Ca by 20%–60% in only 120 years. We estimated unmeasured Ca inputs from rock breakdown, root-zone deepening, and dry deposition; these should not be expected to make up the Ca deficit. Acid precipitation may be the cause of current high leaching of Ca. Although Ca deficiency does not generally occur now in acid forest soils, it seems likely if anthropogenic leaching and intensive harvest removal continue.

The relationship of nutrient depletion to long-term productivity of forests depends on the total amount of each nutrient in the ecosystem as well as on the availability of that nutrient. The question addressed by this paper is: do the total amounts of nitrogen, calcium, potassium, magnesium, and phosphorus in a forest ecosystem change significantly over time? Only input and output rates and the total pool size of each nutrient are needed to answer this question. The complexities of various nutrient fluxes within the system and of the availability of nutrients to plants are avoided. Continual depletion of the total amount of any nutrient must sooner or later decrease its availability and, consequently, forest productivity.

Both timber harvest and leaching remove nutrients from forest ecosystems. Since the early 1970s research has tried to quantify these removals and to assess their effects.

Nitrogen often limits forest growth. High nitrification and N leaching after harvest on some sites (Bormann and others 1968) contributed to concern about N depletion. Ecosystem simulations by Aber and others (1978) and Waide and Swank (1977) suggested

that intensive harvesting could cause serious N deficiency in short rotations.

However, most studies of nutrient removal by whole-tree harvest in the northeastern United States and eastern Canada have found that Ca was the element most likely to be depleted (Weetman and Weber 1972, Boyle and others 1973, Silkworth and Grigal 1982). Mann and others (1988) summarize a number of studies nationwide, all of which indicate a potential problem with Ca. In these and other studies (White 1974, Jorgenson and others 1975), inadequate knowledge of atmospheric inputs, imprecise estimates of leaching losses, lack of measurements of total pool sizes, and questions about weathering rates result in considerable uncertainty about interpretation.

In this paper we analyze the only studies from the eastern United States that we know of in which input, leaching, harvest removal, and total soil and biomass pools have all been measured. These studies cover six sites in Tennessee and New England, including both residual and glaciated soils and both coniferous and hardwood forests (Table 1). At each site, the biomass and soil pools of nutrient elements, input from the atmosphere, output in leaching or streamflow, and harvest removal by clear-cutting have been previously reported. Starting with these published values, we use a linear extrapolation over time to show the expected

KEY WORDS: Acid precipitation; Biomass nutrients; Calcium; Clear-cutting; Magnesium; Nitrogen; Phosphorus; Potassium; Soil leaching; Soil nutrients; Timber harvest; Weathering; Whole-tree harvest

changes in total pool sizes over 120 years. For Ca, which depletes rapidly, we estimate additional unmeasured inputs by dry deposition, rock breakdown (weathering), and root-zone deepening.

Methods

Study Sites and Methods

The basic data used in this paper have been published by Johnson and Todd (1987) and Hornbeck and others (1987, 1989). Methods varied somewhat among these studies. Here we give only a general overview of them; the original sources should be consulted for details (Table 1).

In this paper the *total* pool of a nutrient is defined as the total mass of the element in all the organic matter on the site and in the mineral material <2 mm in particle size down to the bottom of the root zone. Gravel and stones >2 mm in diameter are excluded from the total pool. We call this material *exclusions*. The total nutrient pool does not include the mass of nutrient elements in material >2 mm, that is, in the exclusions. This point will be important to later discussion.

Statistical error was not evaluated in the original papers for the data we use. In our analysis, each of the six stands is a sample (but not random) of the population of all forests on acid soils in the eastern United States. There is general agreement in our results among sites; averaging over sites or statistical analysis for differences among sites is not warranted.

At each of the study sites, allometric equations relating tree biomass to tree size were combined with sampled nutrient concentrations in the biomass to calculate the nutrients in above ground biomass. At some sites, nutrients in root biomass were estimated proportional to the above ground pool. Nutrient contents of forest floor or O horizon were obtained by direct sampling. For the mineral soil, samples were collected from soil pits, by horizon or depth. All samples were sieved to 2 mm and then analyzed for nutrient concentrations by methods giving the *total* amount of that element in the <2-mm sample. Total pool size for the mineral soil was calculated as the sum over horizons or depths of the product of bulk density (<2-mm basis), layer thickness, and concentration by mass. The depth of sampling varied among sites but was always less than 1 m and encompassed most or all of the root zone.

Chemistry of input in precipitation and output in leaching or streamflow was measured frequently over several years at each site. Precipitation input included dry deposition that fell into the chemical sampling

container. Leaching output from mature forest was calculated as the product of either soil–water or stream concentration and an estimated or measured soil–water flux or streamflow. Nutrient leaching increases following harvest on many sites (Mann and others 1988). This harvest-induced leaching loss, an addition to the loss from the mature forest, was calculated similarly using measurements from harvested areas.

At five of the six sites, harvest removal was by whole-tree clear-cutting in which most of the above ground biomass was removed. Biomass removal was measured by direct weighing of trees on sample areas. Whole-tree clearcutting was used because it is both a frequent management practice and because it has a severe impact. At the Hubbard Brook Experimental Forest site (HB), harvest was by a progressive strip-cut leading to a clear-cut after four years. To make the Hubbard Brook data comparable with the other sites, we assume that a whole-tree harvest would have removed 90% of the above ground nutrients. This is in line with the other sites.

We estimate percent change in total pool size of a nutrient over 120 years, C , as

$$C = 100 [120 (I - O) - n (R + L)]/T$$

where I is annual input and O is annual leaching output from uncut forest, R is harvest removal, L is additional harvest-induced leaching, T is the initial total pool, and n is the number of harvests in the 120-year period. We evaluated the results for $n = 0$, $n = 1$, and $n = 3$. With $n = 0$, there is no harvest and the current uncut input–output condition is assumed to continue through the whole 120 years. A single harvest every 120 years ($n = 1$) is a reasonable *long rotation* for the northeastern United States (Likens and others 1978). Three harvests every 120 years ($n = 3$) is a reasonable *short rotation*.

We recognize that harvest removals in the second and third cuts of the short rotation will not equal the removal in the first cut of an older forest because regrowth will not have replaced the total biomass. Using the same R for all three cuts thus slightly overestimates C when $n = 3$, but this is not important to our general conclusions.

Results

Pool Sizes

The size of any given pool of a nutrient element generally differs by a factor of 3 or less among the six mature stands (Table 2). The largest difference in above ground pools occurs for Ca in Tennessee (MO and LP), where the mixed oak contains 1290 kg/ha

Table 1. Characteristics of the study sites.

Site	Location	Latitude	Longitude	Forest type	Principal species	Principal soil types
ME ^a	T4R12, ME	45°57'N	69°19'W	spruce–fir	<i>Picea rubens</i> Sarg., <i>Abies balsamea</i> (L.) Mill.	coarse-loamy, mixed, Aquic Haplorthod, and Aeric Haplaquept
NH ^b	Success, NH	44°30'N	71°2'W	northern hardwood	<i>Acer saccharum</i> Marsh., <i>Betula alleghaniensis</i> Britt., <i>Fagus grandifolia</i> Ehrh.	coarse-loamy, mixed, frigid, Typic Haplorthod
HB ^c	Woodstock, NH	43°57'N	71°42'W	northern hardwood	<i>Acer saccharum</i> Marsh., <i>Betula alleghaniensis</i> Britt., <i>Fagus grandifolia</i> Ehrh.	coarse-loamy, mixed, frigid, Typic Haplorthod
CT ^d	Chester, CT	41°24'N	72°31'W	central hardwood	<i>Quercus</i> spp., <i>Carya</i> spp., <i>Acer rubrum</i> L., <i>Betula lenta</i> L.	coarse-loamy, mixed, mesic, Typic Dystrochrept
MO ^e	Oak Ridge, TN	35°58'N	84°20'W	central hardwood	<i>Quercus</i> spp., <i>Liriodendron tulipifera</i> L., <i>Acer rubrum</i> L., <i>Carya</i> spp.	clayey, kaolinitic, thermic, Typic Paleudult
LP ^e	Oak Ridge, TN	35°58'N	84°20'W	loblolly pine	<i>Pinus taeda</i> L.	clayey, kaolinitic, thermic, Typic Paleudult

^aSmith and others 1986, Hornbeck and others 1989.

^bHornbeck and Kropelin 1982, Hornbeck and others 1989.

^cLikens and others 1977, Hornbeck and others 1987, Huntington and others 1988.

^dTritton and others 1987, Hornbeck and others 1989.

^eJohnson and Todd 1987.

while the loblolly pine has only 200 kg/ha. Above-ground, root, and forest floor pools are much smaller than the mineral soil pools except for relatively large amounts of forest floor N in the four New England sites. In the mineral soil, the two Tennessee sites have high K and the Maine (ME) site has high Mg. For the purposes of this study, however, one should focus on the general similarity of pool sizes among sites rather than the few differences.

Input–Output

Annual inputs of nutrients in precipitation and outputs in streamflow or leachate for mature forests are generally consistent over the region (Table 3). For most elements at most sites, inputs are less than or equal to 2 kg/ha/yr. N input is 4–11 kg/ha/yr throughout the region, while Ca input in Tennessee is 6–7 kg/ha/yr. On the output side, Ca consistently has the highest mass losses, followed successively by Mg, K,

N, and P. Negligible amounts of P move either in or out of all systems. Ca output exceeds input by 4–16 kg/ha/yr. On the other hand, N input exceeds output by 2–8 kg/ha/yr. Both K and Mg generally have net losses of several kilograms per hectare per year.

Harvest Removal and Induced Leaching

Whole-tree harvest generally removed about 90% of the above ground pool of each nutrient (Tables 2 and 3).

The amount of additional leaching loss induced by harvest varied (Table 3). No data were presented for the Tennessee sites (MO and LP), but postharvest NO₃⁻ leaching was very low at these sites (Johnson and Todd 1987), so the additional leaching loss for each element probably was also low. Data for the Hubbard Brook (HB) harvest extended over ten postharvest years as opposed to three years for the other sites. On the other hand, the Hubbard Brook harvest was

Table 2. Component pools and total (<2 mm) pool of N, Ca, K, Mg, and P for six mature forests.

Site	Component (kg/ha)				Total
	Above ground	Roots	Forest floor	Mineral soil	
Nitrogen					
ME	410	140	920	5830	7300
NH	260	190	1740	4800	6990
CT	300	120	1000	3600	5020
HB	350	180	1300	5900	7730
LP	200	—	440	3050	3690
MO	420	—	150	3160	3730
Calcium					
ME	540	190	380	10330	11440
NH	360	120	490	7570	8540
CT	590	240	100	3320	4250
HB	380	100	370	9600	10450
LP	200	—	160	5130	5490
MO	1290	—	160	4860	6310
Potassium					
ME	240	80	70	10000	10390
NH	140	70	80	5080	5370
CT	180	70	70	5040	5360
HB	160	60	70	5080	5370
LP	120	—	40	23200	23360
MO	190	—	20	23000	23210
Magnesium					
ME	60	20	70	36450	36600
NH	40	20	130	7660	7850
CT	40	20	50	13290	13400
HB	40	10	40	7700	7790
LP	50	—	20	6300	6370
MO	50	—	20	7350	7420
Phosphorus					
ME	60	20	80	2700	2860
NH	20	20	90	1140	1270
CT	20	10	40	970	1040
HB	40	30	60	2520	2650
LP	20	—	30	790	840
MO	30	—	10	1000	1040

less intense than a whole-tree harvest. At Hubbard Brook, N, Ca, and K losses exceed those from the other sites.

The harvest removal plus harvest-induced leaching for a single harvest amounted to 3% or less of the total pool of K, Mg, and P at all sites. N loss from harvest was consistent at 4%–8% of the total pool. Ca loss for non-oak forests represented 2%–5% of the total pool, but for CT and MO forests, which contain oak and hickory, the loss was 13% and 19%, respectively, of the initial Ca pool.

120-Year Extrapolation

The estimated percent changes in total pool sizes of each nutrient over 120 years, *C*, are fairly consistent

Table 3. Annual input and output, whole-tree harvest removals, and harvest-induced leaching losses for six forests.

Site	Rate (kg/ha/yr)		Amount (kg/ha)	
	Annual input	Annual output	Harvest removal	Harvest-induced leaching
Nitrogen				
ME	4	0	380	6
NH	6	3	240	6
CT	8	0	270	19
HB	6	4	315	58
LP	7	2	200	—
MO	11	3	310	—
Calcium				
ME	1	16	500	43
NH	1	15	340	30
CT	2	10	530	28
HB	2	12	340	48
LP	6	22	200	—
MO	7	11	1090	—
Potassium				
ME	0	2	220	29
NH	1	2	130	5
CT	5	5	160	23
HB	1	2	140	48
LP	1	5	120	—
MO	2	5	120	—
Magnesium				
ME	0	5	50	13
NH	0	4	40	12
CT	1	7	40	16
HB	1	3	40	7
LP	1	4	50	—
MO	1	6	40	—
Phosphorus				
ME	0	0	50	0
NH	0	0	20	0
CT	0	0	20	0
HB	0	0	40	0
LP	0	0	20	—
MO	0	0	20	—

among sites (Table 4). Because input of nitrate from acid deposition is high, ecosystem N tends to increase over time except with three harvests, for which the system is balanced or has some loss. For K, Mg, and P, depletions are small (0%–3%) to moderate (5%–10%) in all cases, with greater depletion expected with three harvests. However, Ca depletion is expected to be large on all sites: 8%–35% over 120 years in the absence of any timber harvest, and 23%–62% with three whole-tree harvests.

Discussion

Forest harvesting removes nutrients from the system in larger quantities but at infrequent intervals

Table 4. Calculated percent change in total nutrient pool for no, one, and three whole-tree harvests over 120 years.

Site	Pool change (%) over 120 years		
	No harvest	1 harvest	3 harvests
Nitrogen			
ME	7	1	-9
NH	5	2	-5
CT	19	13	2
HB	3	-2	-11
LP	16	11	0
MO	26	17	1
Calcium			
ME	-16	-20	-30
NH	-20	-24	-33
CT	-23	-36	-62
HB	-11	-15	-23
LP	-35	-39	-46
MO	-8	-25	-59
Potassium			
ME	-2	-5	-9
NH	-2	-5	-10
CT	0	-3	-10
HB	-2	-6	-13
LP	-2	-3	-4
MO	-2	-2	-3
Magnesium			
ME	-2	-2	-2
NH	-6	-7	-8
CT	-5	-6	-7
HB	-3	-4	-5
LP	-6	-6	-8
MO	-8	-9	-10
Phosphorus			
ME	0	-2	-5
NH	0	-2	-5
CT	0	-2	-6
HB	0	-2	-5
LP	0	-2	-7
MO	0	-2	-6

compared to the continual leaching loss. Over a 120-year period, leaching and harvest removal contribute roughly equally to nutrient depletion. Both must be considered in analyses of long-term productivity (D. W. Johnson and others 1988).

Consistency in above ground pool size of each element among sites and with other temperate forests of similar age (Morrison and Foster 1979) implies that harvest removals can be estimated easily. In the north-eastern United States, a whole-tree harvest of a mature forest will remove roughly 250 kg/ha of N, 350 kg/ha of Ca (600 for oak-hickory forest), 150 kg/ha of K, 40 kg/ha of Mg, and 20 kg/ha of P. Stem-only harvest would remove about half of these amounts. The effect of stem-only rather than whole-tree harvest is not

shown here but can be approximated from the *C* equation by using half the *R* values given in Table 3.

Input of nitrates from acid precipitation exceeds leaching losses and can replace the N removed by long-rotation harvests on all sites. With intensive harvest at short rotations, some sites appear susceptible to N depletion (Aber and others 1978). Gaseous exchanges of N, including dry deposition of nitric acid, nitrogen fixation, and denitrification are poorly quantified for forests, but there is no evidence for substantial changes of total N by such processes.

For P and K, harvest removals generally exceed the net leaching loss in the absence of harvest, whereas for Mg the reverse is true. For all three elements, projected long-term depletion of the total pools is cause for some concern, but the depletions are considerably smaller than the depletion expected for Ca.

Calcium depletion by leaching from mature north-eastern forests in the absence of timber harvest is large with respect to the total <2-mm pool (Table 4). Additional Ca removal by three whole-tree harvests in 120 years doubles the expected depletion rate. Johnson and Todd (1987) reached a similar conclusion for their Tennessee sites. Calcium is not normally considered to be a limiting nutrient for forest growth in the eastern United States. However, our projected rapid depletion from leaching and intensive harvest suggests that Ca deficiency could become a problem in the future.

Such a nutrient depletion scenario is usually countered by invoking various inputs not included in our analysis. These are dry deposition, weathering, and root-zone deepening. We now discuss each of these quantitatively.

Dry Deposition and Gaseous Fluxes

Bulk precipitation collectors are open to the atmosphere continually and thus include both wet deposition and some dry deposition. Bulk collectors were used in all the studies we summarized. Dry deposition directly to the forest canopy may differ from that to a collector but has proven extraordinarily difficult to measure (Hicks 1986). For N, additional dry deposition to the canopy may be as much as the bulk collection (D. W. Johnson and others 1985). For Ca at Oak Ridge, D. W. Johnson and others (1985) estimated the sum of wet and dry deposition to the canopy to be 60% greater than the bulk precipitation value; this gave additional dry deposition of Ca as 3.2 kg/ha/yr. For the New England sites, and for the Northeast as a whole (NAPAP 1987), Ca input is only 1–2 kg/ha/yr. Even doubling this annual input rate to account for

additional dry deposition has a rather small effect on the rate of Ca depletion.

Weathering

Weathering implies conversion of minerals from stable forms, as in rock or soil minerals, into forms that can participate in a variety of reactions involving exchange, leaching, and biological uptake. But when considering the *total* pool of a nutrient, it is important to differentiate between weathering as an internal flux within the total pool and weathering as an input flux that contributes additional nutrients to the pool. Many papers, such as that of Li and others (1988), which discuss mineral weathering and soil acidification, deal only with internal processes involving <2-mm particles. Such weathering is not an input to the total pool.

Weathering often is calculated by the cation-budget method, which assumes a constant amount of *exchangeable* cation within the system (N. M. Johnson and others 1968, Reuss and Johnson 1986). At Hubbard Brook, Ca weathering calculated by this method is 21 kg/ha/yr (Likens and others 1977). However, if this amount comes from the dissolution of mineral particles <2 mm, it is an internal process that does not increase the amount of total Ca.

The only form of weathering that contributes input to the total pool is the reduction of particle size from >2 mm to <2 mm in diameter. To avoid confusion with other usages of the term weathering, we call this *exclusion breakdown*.

Exclusion breakdown is related to the rate of formation of weathering rinds on rock (Clayton 1979). Birkeland (1984) summarized several studies of weathering rind formation, giving average rates of 0.1–0.2 mm in 10,000 years and maximum rates of 3–6 mm in 10,000 years for the most highly weatherable rock types. At Hubbard Brook (HB), bedrock is virtually unweathered where the till is more than 1.5 m thick (N. M. Johnson and others 1968). Glacial striations over 10,000 years old appear to have weathered only 1 or 2 mm.

Calcium content of local rocks at Hubbard Brook is roughly 1% (N. M. Johnson and others 1968, England 1976). With a rock density of 2650 kg/m³, this gives a Ca content of 265 kg/ha/mm of rock. In a 70-cm root zone, stone content of 20% by volume then means that 37,000 kg/ha of Ca is stored in the exclusions. This is a large amount compared to the <2 mm pool size. The question becomes the rate at which breakdown of stone exclusions occurs.

Complete breakdown of a 1-mm weathering rind in 10,000 years provides only 0.026 kg/ha/yr of Ca from rock of 1% Ca. The total exclusion breakdown rate

depends on the number of rinds or the amount of stone surface per unit area of land. If the size-class distribution of >2-mm particles is known, the effective number of weathering rinds can be calculated. The surface-to-volume ratio of a sphere of diameter d is $6/d$. For a given volume fraction, v , of rock of diameter, d , over a soil thickness, t , the surface area of rock per unit surface area of land is $6 vt/d$. For instance, 5% by volume of 5-mm gravel plus 15% by volume of 5-cm stones in a 70-cm root zone gives 42 plus 13 = 55 effective weathering rinds. If each rind weathers at 1 mm in 10,000 years, then $55 \times 0.026 = 1.5$ kg/ha/yr of Ca is an estimate of exclusion breakdown. There is obviously great uncertainty in this number, but better estimates are not yet available.

In residual soils, such as the Tennessee sites, exclusion breakdown has been proceeding for millions of years. Bedrock normally is far below the root zone and coarse fragments often are small and composed primarily of chert, which has no calcium. The presence of any coarse fragments at all in the solum of old soils not affected by erosion implies that the rate of exclusion breakdown is very slow. Exclusion breakdown cannot provide substantial Ca input at the Tennessee sites.

Root-zone Deepening

Gradual deepening of the lower boundary of the ecosystem (the root zone) would represent an input to the system and occurs when roots exploit new volumes of parent material not previously available to them. When the lower boundary is at the bedrock surface, weathering of the bedrock surface is effectively a deepening of the lower boundary, and is a system input. As was shown, the contribution of a bedrock weathering rind of 1 mm in 10,000 years is negligible with respect to measured inputs and outputs of Ca.

Root-zone deepening in the absence of bedrock can occur as a long-term process only if the soil surface also is being lowered. On a time scale of thousands of years, the root-zone thickness of a forest must be virtually constant. Therefore, root-zone deepening can occur only at a rate equal in mass to the rate of net particulate and dissolved mass outflow from the system.

In nonerosive forest systems, particulate output is small. At Hubbard Brook, particulate mass output in first- and second-order streams is only 25 kg/ha/yr compared to dissolved output of 140 kg/ha/yr (Bormann and others 1969). The particulate output value is confirmed by sedimentation studies in nearby Mirror Lake, which suggest that this rate has been stable for the 10,000 years since postglacial reforestation (Davis and others 1985). If the Ca content of this

material is the same as the local rocks, about 1%, the particulate Ca outflux is 0.3 kg/ha/yr.

The total particulate and dissolved outflow rate at Hubbard Brook is equivalent in mass to only 6 cm of rock thickness in 10,000 years. At a soil bulk density of 1.0 Mg/m³, this is equivalent to only 16 cm of soil. This implies that the soil surface has been nearly stable since postglacial reforestation. The following independent argument supports this stability.

Many till soils in the glaciated northeastern United States have a dense C horizon beginning at a soil depth of about 70 cm. The genesis of this material has been controversial; it has been called either a pedogenic fragipan or a glacially compacted till (Grossman and Carlisle 1969). If it is a product of subglacial compaction, with the 70-cm root zone above it produced by periglacial and postglacial processes, then the root zone in such soils has been stable for most of the 10,000 years since glacial retreat. A possible 16-cm change is not inconsistent with this.

At 6 cm of rock thickness in 10,000 years with 265 kg/ha/mm of Ca, or with 16 cm of soil with an equivalent amount, the Ca input from root-zone deepening is 1.6 kg/ha/yr. Subtracting 0.3 kg/ha/yr for particulate output in streamflow gives a net particulate input from root-zone deepening of 1.3 kg/ha/yr. This is insufficient to alleviate the present input–output imbalance of Ca, but further refinement of the estimate should be attempted.

We believe that most northeastern forests behave similarly to Hubbard Brook with respect to mass loss and root-zone deepening. In more highly erosive systems, particulate movement by mass flow, erosion, and stream particulates may be considerably higher. Depending on the site, the net change in mass may be positive (deposition) or negative (erosion), the root zone moving either upward or downward, respectively. In either case, the nutrient budgets could be altered considerably from the examples presented here.

Cation Depletion and Acid Precipitation

The reported differences between input and output of Ca, Mg, and K (Table 3) are so great that current rates would deplete the total <2-mm pool of these elements in several thousand years or less at most of the sites in this study. This is true even for unmanaged forests without timber harvest. For Ca, all sites would be completely depleted in less than 1000 years in the absence of exclusion breakdown, root-zone deepening, or additional dry deposition. In the literature, dry deposition and weathering usually have been invoked to make up such deficits. Our analysis of Ca suggests very approximate values of 2.0, 1.5, and

1.3 kg/ha/yr of Ca for inputs by additional dry deposition, exclusion breakdown, and root-zone deepening, respectively. These add to about half of the reported input–output imbalance in Ca in the absence of harvest removal.

The values presented in Tables 2–4 are measured or calculated with reasonable accuracy. General agreement among sites and among investigators provides some confidence in the values. By contrast, the additional inputs from dry deposition, exclusion breakdown, and root-zone deepening are little better than educated guesses. In terms of ecosystem preservation and maintenance of long-term productivity, it seems prudent *not* to rely on such poorly evaluated inputs to offset the well-documented rates of nutrient depletion.

The current input–output imbalances cannot have been maintained for long or the basic elements already would be depleted. Current rates of cation depletion in the northeastern United States probably are the result of anthropogenic acid precipitation (D. W. Johnson and others 1988), and the net loss may only have existed over roughly 100 years of the Industrial Age. In much of the Northeast, precipitation pH averages as low as 4.0, whereas “clean” or preindustrial precipitation probably had a pH above 5.0 (Galloway and others 1984), so the contribution of anthropogenic acidity is about 1 kmol(+) /ha/yr. One kmol (+) is 1000 equivalents of positively charged ions. As water passes through the forest ecosystem, the H⁺ in solution is replaced by the base cations Ca²⁺, Na⁺, Mg²⁺, and K⁺ (Table 5) (Reuss and Johnson 1986, Driscoll and others 1988). If preindustrial H₂SO₄ and HNO₃ inputs were one tenth of current rates, base-cation outputs also may have been on the order of one tenth of current rates. Assuming unchanged base-cation inputs, outputs would then have been equal to or less than inputs (Table 5) and the system would have been in long-term balance with respect to the basic cations.

Conclusions

Studies of six different forests suggest that calcium is being depleted by leaching from eastern forest soils at a rate that will remove all <2-mm ecosystem Ca in less than 1000 years. This excessive leaching rate cannot have continued for thousands of years and may have been induced by recent acidification of precipitation (D. W. Johnson and others 1988). Removal of Ca in above ground biomass by timber harvest can double this net loss of Ca. The combination of leaching loss and whole-tree harvest at short (40-year) rotations apparently could remove roughly 50% of biomass and soil Ca in only 120 years.

Table 5. Mean annual ionic input in precipitation and output in streamflow at the Hubbard Brook Experimental Forest, 1963–1974^a

	Cations [kmol(+)/ha/yr]			Anions [kmol(–)/ha/yr]	
	Input	Output		Input	Output
H ⁺	0.97	0.10	SO ₄ ^{2–}	0.80	1.12
NH ₄ ⁺	0.16	0.02	NO ₃ [–]	0.32	0.28
Ca ²⁺	0.11	0.68	Cl [–]	0.20	0.14
Na ⁺	0.07	0.32	HCO ₃ [–]	0.00	0.13
Mg ²⁺	0.05	0.26			
K ⁺	0.02	0.05			
Al ³⁺	0.00	0.21			
Total	1.38	1.63		1.32	1.66

^aLikens and others 1977.

Unmeasured Ca inputs from dry deposition, rock weathering, and root-zone deepening have been estimated here, but the uncertainties in the estimates are large. Further research is needed. In the meantime, these inputs should not be counted on to counteract the measured depletion rates.

The nutrients Mg and K are also subject to depletion by leaching and harvest removal, although not as severely as Ca. Phosphorus is tightly held and is not likely to be depleted. Nitrogen is replaced by anthropogenic atmospheric deposition and is only subject to depletion on some sites with intensive harvest.

Calcium deficiency is rarely mentioned as a limiting factor in growth of northeastern forests. If Ca depletion is a recent phenomenon, Ca deficiency may not yet have had time to develop. Available or exchangeable Ca could be depleted at a faster rate than total Ca, and cation imbalances also may be induced. Shortle and Smith (1988) have proposed aluminum-induced calcium deficiency as a cause of high mortality of red spruce (*Picea rubens* Sarg.) in the mountains of the northeastern United States. This could be the first indication of Ca deficiency.

Three measures are possible to mitigate Ca depletion: reduction of acid deposition to preindustrial levels, restrictions on short-rotation whole-tree harvesting, and liming of vast forest areas on a scale similar to liming agricultural crops. About the first, enough has been said elsewhere. The second is only a partial solution or a delaying tactic. Large-scale liming, perhaps at the same time as harvest, may be required if forest productivity is to be maintained at present levels.

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